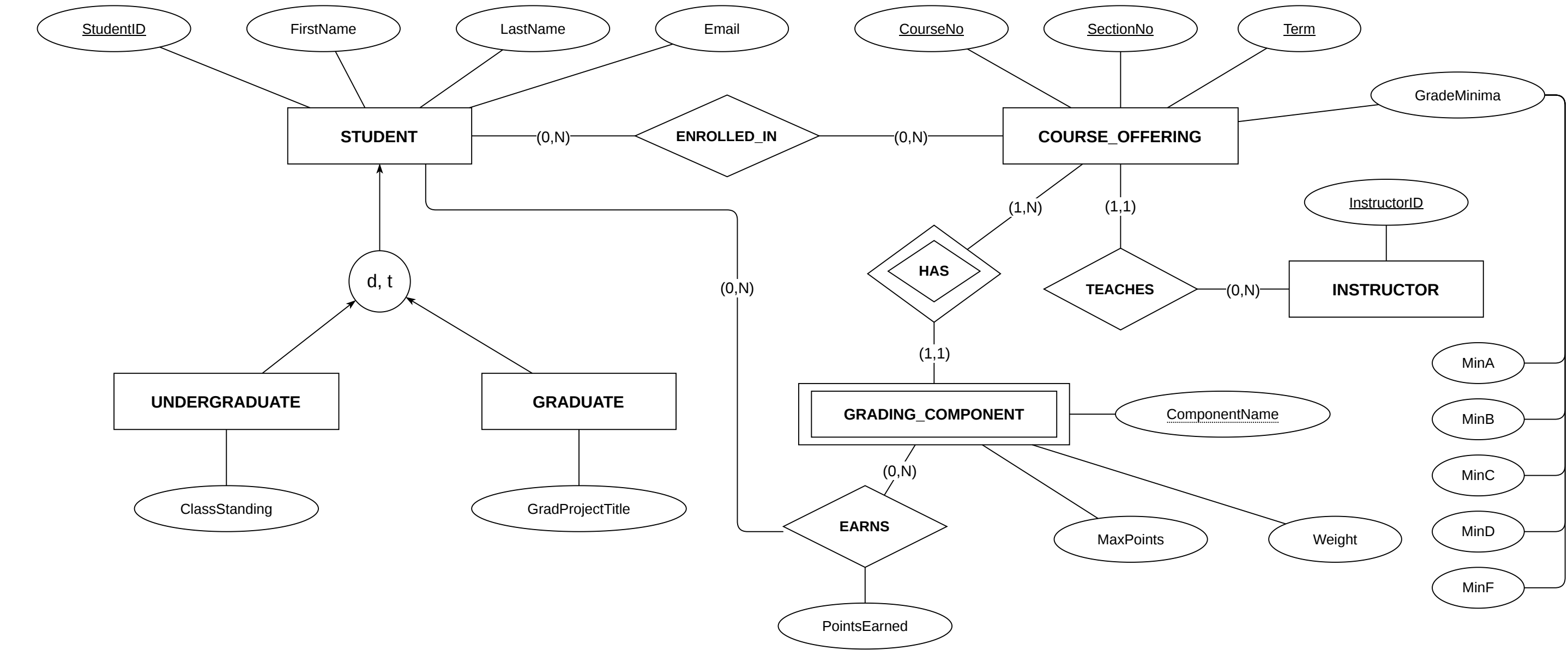


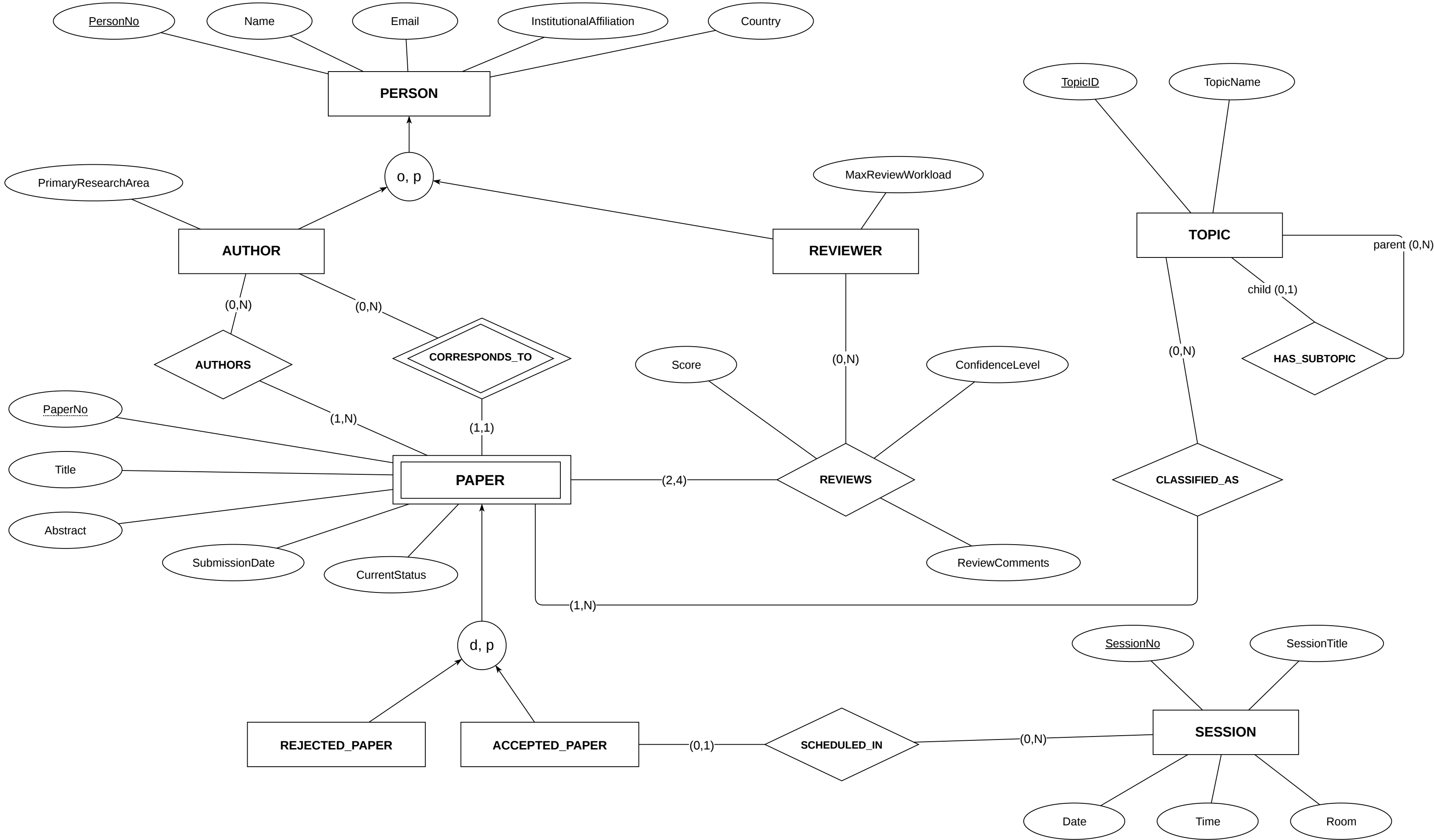
Ans 1 : GRADE_BOOK - Enhanced Entity-Relationship Diagram



CONSTRAINTS, KEYS, AND ASSUMPTIONS

1. **Subclass constraints:** d = disjoint; t = total. Every student is exactly one of UNDERGRADUATE or GRADUATE. Both inherit StudentID and all student attributes.
2. **Keys:** STUDENT: StudentID; INSTRUCTOR: InstructorID; COURSE_OFFERING: (CourseNo, SectionNo, Term). Term includes the year.
3. **Weak entity:** GRADING_COMPONENT is shown with a double rectangle. Its owner is COURSE_OFFERING, and HAS is its identifying relationship. ComponentName is its partial key, shown with a dotted underline. Its full key is (CourseNo, SectionNo, Term, ComponentName).
4. Each offering has exactly one instructor and at least one component. Instructors may have no offerings. Students and offerings may have no enrollments.
5. Each student-offering enrollment pair is unique. EARNs stores at most one PointsEarned value per student-component pair.
6. A score exists only if the student is enrolled in the offering that owns the component. Ungraded components have no EARNs record yet; both EARNs minima are 0.
7. Assume no extra credit: MaxPoints > 0; 0 <= PointsEarned <= MaxPoints; 0 <= Weight <= 100. Weights normally total 100%, but this is not a strict constraint.
8. ClassStanding is freshman, sophomore, junior, or senior. GradeMinima is composite, containing MinA, MinB, MinC, MinD, MinF; assume MinA > MinB > MinC > MinD >= MinF.
9. Each (min,max) label counts the adjacent entity's participation. Underline = key; dotted underline = partial key; double rectangle = weak entity; diamond = identifying relationship.

Ans 2 : CONFERENCE_REVIEW - Enhanced Entity-Relationship Diagram



CONSTRAINTS, KEYS, AND ASSUMPTIONS

- PERSON specialization: o, p.** Overlapping and partial: a person may be an author, a reviewer, both, or neither. Both subclasses inherit PersonNo and all PERSON attributes.
- PAPER specialization: d, p.** ACCEPTED_PAPER and REJECTED_PAPER are disjoint and partial; papers awaiting a decision belong to neither. Both inherit PAPER's full key and attributes.
- Weak entity:** PAPER is shown with a double rectangle. Its owner is its corresponding AUTHOR; CORRESPONDS_TO is the identifying relationship. PaperNo is the dotted-underlined partial key.
- Keys:** PERSON = PersonNo; PAPER = (CorrespondingAuthorPersonNo, PaperNo); TOPIC = TopicID; SESSION = SessionNo. TopicID is an assumed unique identifier; topic names need not be unique.
- Authorship constraint:** Every paper has at least one author and exactly one corresponding author. Every CORRESPONDS_TO pair must also occur in AUTHORS. Registered authors may have no papers yet.
- Topic hierarchy:** Each child topic has at most one immediate parent; each parent may have many immediate children. The hierarchy is acyclic, with no self-parent links; multiple root topics are allowed.
- Every paper has one or more directly assigned topics. Topics may have no papers. Ancestor classifications can be inferred through the hierarchy; they need not be stored as additional assignments.
- Reviews:** Each paper has 2–4 distinct assigned reviewers. Each reviewer-paper pair occurs at most once. Assignment count per reviewer must not exceed that reviewer's nonnegative integer MaxReviewWorkload.
- Score, ConfidenceLevel, and ReviewComments belong to the REVIEWS assignment. They may be absent while a review is pending and are recorded on completion. Score and confidence scales are conference-defined.
- The 2–4 reviewer constraint models the required assigned state. Assume reviewers cannot review papers they author; this is an additional conflict-of-interest policy, not a stated requirement.
- CurrentStatus is consistent with subclass membership: accepted iff in ACCEPTED_PAPER; rejected iff in REJECTED_PAPER. Other statuses indicate neither subtype. Only accepted papers may be scheduled.
- Accepted papers may be unscheduled; sessions may initially be empty. Each accepted paper belongs to at most one session. This database covers one conference; no separate conference entity is assumed.
- Notation:** Each (min,max) counts the adjacent entity's participation. Underline = key; dotted underline = partial key; double rectangle = weak entity. Parent/child labels distinguish recursive roles.